Arushan Sinnadurai

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EDUCATION

Master of Biotechnology September 2025 – May 2027
Carleton University, Ottawa ON
GPA 4.0

Bachelor of Engineering, Biomedical and Electrical Engineering September 2016 – May 2021
Carleton University, Ottawa ON
GPA 3.2

PROFESSIONAL SUMMARY

Biomedical engineer and software developer with 5+ years of experience building distributed data systems and backend applications. Currently pursuing a Master of Biotechnology to bridge software engineering and life sciences. Passionate about leveraging data engineering, bioinformatics, and AI to accelerate biotech and healthcare innovation.

WORK EXPERIENCE

Software Engineer, Data Operation April 2026 – Present
Flosonics Medical, Toronto, ON

- Built and maintained **DBT and Apache Airflow data models and pipelines**, supporting reliable and scalable data transformation workflows.
- Improved workflow efficiency by **40%** through process optimization and enhancements to **ground-truth labeling** workflows.
- Developed meaningful and insightful **Tableau dashboards**, enabling stakeholders to monitor key metrics and derive actionable insights from data.

Data Engineer July 2021 – April 2026
8x8 Inc, Ottawa, ON

- Developed **Java Spring Boot** applications and **Apache Flink** pipelines in **Scala** capable of ingesting **100,000+ records** for real-time and batch ETL processing.
- Built and operated distributed systems for data ingestion, extraction, and transformation of multi-terabyte datasets.
- Created a **Java Spring Boot microservice** for optimized data retrieval from PostgreSQL, reducing search query times by **20%**.
- Collaborated with cross-functional teams to integrate machine learning outputs into production data streams.

Software Developer Intern January 2026 – April 2026
CardiAxis Medical, Ottawa, ON

- Developed scalable **polar map visualization** algorithms for cardiac imaging, enabling dynamic ring-based sampling beyond the standard 16-ring resolution to accommodate variable LV geometry.

- Engineered **3D surface object processing pipelines** to convert vertex-based scalar data (ECGi, PET, wall thickness, scar tissue) into polar map representations with temporal dimension support.
- Built interactive **target selection tools** with spray-paint contour manipulation, converting 2D polar map selections into 3D contours for myocardium segmentation.
- Implemented **Risk Traceability and QMS Methodology** for medical device software to ensure regulatory compliance and code quality standards.

Software Developer Intern September 2019 – September 2020
Kinaxis, Ottawa, ON

- Implemented **automated test suites (JUnit)**, improving coverage by **30%** and ensuring robust code reliability.
- Developed **microservices** to efficiently migrate large-scale datasets via **Kafka**, enhancing data throughput.
- Modernized deployment pipelines by introducing **Docker** and **Helm 3**, improving release frequency and scalability.

Software Implementation Support May 2019 – September 2019
University of Ottawa Heart Institute, Ottawa, ON

- Supported the rollout of the **EPIC Electronic Health Information System**, providing technical support and troubleshooting.
- Collaborated with healthcare teams to optimize workflows for cardiac imaging registration and documentation.
- Analyzed data integration issues and developed best practices for diagnostic imaging data management.

TECHNICAL SKILLS

- Languages: Python, Java, Scala, JavaScript (React), SQL, C++
- Data & Infrastructure: Apache Kafka, Apache Flink, Spring Boot, ETL Pipelines, PostgreSQL, Docker, Helm, AWS, Git
- Medical Software: 3D Slicer, Cardiac Imaging, Polar Map Visualization, QMS Methodology, Risk Traceability, IEC 62304
- Domains: Biotechnology, Biomedical Systems, Bioinformatics, Machine Learning, Health Informatics